Pancreatic Cancer Prognosis Using Clinical and Radiomic Data

UE24CS352A — Machine Learning Mini-Project | Draft pending dataset and results

1. Problem Statement

This project explores whether clinical information and quantitative radiomic features from medical images are associated with pancreatic cancer survival outcomes. It follows the direction of the Stanford CS229 reference report by Jamalian (2020), which compares clinical, radiomic and combined features using survival analysis.

2. Dataset

The reference describes 80 patients and 900 MRI-derived radiomic features per patient, plus clinical variables. It states that the original Stanford Cancer Center dataset was supplied under a non-disclosure agreement and is not included with the paper. This project therefore requires a separately verified public or authorized, de-identified dataset containing survival/follow-up duration, event status (observed event versus right-censoring), and compatible predictors. The final report must document the actual collection, usable case count, event count, missing values and feature source. The included CSV template is a schema only, not data.

3. Methodology

The planned workflow validates outcome fields, preprocesses clinical and radiomic predictors, and fits a penalized Cox proportional hazards model. The concordance index can assess ranking of survival outcomes while accounting for censoring. Kaplan–Meier curves can visualize survival among groups defined by predicted risk. Clinical-only, radiomic-only and combined models should be compared only if the dataset supports them. Feature selection and preprocessing must be fitted within training folds to avoid information leakage.

4. Implementation

The Python pipeline is in src/train_model.py. It expects a CSV containing survival_days, event, and one or more feature columns. It creates model coefficients, a summary, a risk-score distribution and Kaplan–Meier curves. The current score is calculated on training data and is optimistic; held-out or cross-validated evaluation is required before reporting generalization performance.

5. Conclusion and Limitations

This package provides a reproducible starting workflow, but final findings depend on verified patient-level data and evaluation. The reference paper's combined-model C-index of 0.82 must not be presented as this project's result. Small cohorts and correlated radiomic features create substantial overfitting risk. This project is educational and is not intended for clinical decisions.

Reference: Jamalian, A. (2020). CS229 Project Final Report: Pancreatic cancer prognosis using clinical and radiomic data. Stanford University. Technical documentation: lifelines survival regression.